

Overview of all architectures

Summary of every architecture trained for beat-by-beat dRR prediction, with the final selection and test result.

Final selection: tcn_lite_ft_int8 - a causal TCN (24 filters, dilations 1/2/4/7, receptive field 29 beats) on the last 30 RR intervals plus 9 engineered features, predicting a correction on top of an adaptive RLS AR(8) forecast. 8,048 MACs per beat and 8.4 KB of int8 weights. It loses to none of the other candidates in the int8 closed loop on validation and passed every rule on the held-out test subjects.

Goal and constraints

- Predict $dRR[n+1] = RR[n+1] - RR[n]$ beat by beat from the last 30 RR intervals, to fill missing beats in real time on a pacemaker (notebooks/correlation_study_v3.ipynb).
- First hardware: STM32H750 (Cortex-M7); the implant will be weaker. Hard budget: at most 10k MACs per beat (3k preferred), int8 weights at most 32 KB, int8 RMSE within 2% of float, MCU-native ops only (Conv1D, SeparableConv1D, Dense, BatchNorm, ReLU, Add, causal padding, GRU), receptive field at most 30.
- Data: RR series in data/series/*.txt, split by subject: 142 training, 16 validation, 10 test subjects (data/subjects_*.json, stratified by age interval). Selection on validation only; the test split was opened once, for the final evaluation.
- Selection rule (user decisions): inside the budget, precision first, decided by the CLOSED loop of the int8 model with paired per-subject Wilcoxon tests (pooled over drop rates or per rate with Bonferroni, never significantly worse at any rate); true ties go to the smaller model. Every model is also checked for RSA fidelity (no invented or displaced respiratory peak).

Architectures compared (best variant of each family)

family	best variant	MACs/beat	int8 size	val open loop	val closed 10%	val closed 80%	test whole-series 80%	document
tcn_lite	tcn_lite_ft	8048	8.4 KB	32.62	25.87	41.66	43.74	08_tcn_lite.pdf
micro_tcn	micro_tcn	2080	2.4 KB	32.93	26.32	41.86	43.84	07_micro_tcn.pdf
gru	gru	1696	1.8 KB	33.90	28.04	45.07	43.84	06_gru.pdf
mlp	mlp	1136	1.2 KB	34.60	27.07	42.25	44.30	04_mlp.pdf
gbdt	gbdt	3000 compares	238 KB (30,500 nodes)	34.19	25.53	43.16	45.85	05_gbdt.pdf
linear	linear	9	10 params	40.46	30.34	43.36	45.67	03_linear.pdf
TCN_MHA (legacy)	tcn_mha_current	2,040,576	75.5 KB	33.72	26.07	43.88	50.11	09_tcn_mha.pdf
baseline	rls_ar	152 FLOPs	-	37.14	28.88	53.73	49.15	02_baselines.pdf

RMSE in ms. **val open loop**: one-step-ahead on the true history, 16 validation subjects. **10-80%**: closed loop (pacemaker use) on validation: that share of beats dropped at random from beat 0, each filled by the model and fed back; RMSE on model-filled beats, first 5000 beats per subject, mean of the seeds of that round (3 seeds from round 2 on). int8 is used where it exists. **RSA invented**: share of subject-runs whose filled series gains an RSA peak the original lacks. **test whole-series 80%**: diagnostic on the 10 test subjects (whole series, seed 7, results/test/ folder); it was not used for any selection.

Final test (held-out test split, run once, 10 subjects, 942,526 windows)

model	open loop	closed 10%	closed 30%	closed 50%	closed 80%
tcn_lite_ft_int8 (deployed)	28.59	28.25	30.20	33.07	41.55
tcn_lite_ft (float)	28.44	28.04	29.96	32.92	41.54
TCN_MHA (legacy)	29.11	28.58	30.63	33.90	42.53
rls_ar	31.15	29.94	32.54	36.09	54.36
persistence	36.80	33.77	36.74	39.37	46.72

- tcn_lite_ft_int8 is better than rls_ar in 10/10 test subjects open loop (-8.2%) and in the pooled closed loop; at 80% dropped it is 9 ms (23.6%) better. The 10% gain is no longer significant on test.
- On the test subjects it invents and displaces no RSA peak at any drop rate; at 80% dropped it damps RSA power, which a feasibility analysis showed is an information limit (the RSA phase cannot be recovered from 20% of the beats), not a model flaw.

- The legacy TCN_MHA is worse in 10/10 subjects open loop, invents RSA peaks, and cannot run on the MCU.

Correlation between the original and the filled series (validation and test subjects, all models)

model	corr RR validation	corr RR test	corr dRR validation	corr dRR test	RMSE filled, validation (ms)	RMSE filled, test (ms)
micro_tcn_dad	0.9386	0.9423	0.281	0.272	49.10	42.95
tcn_lite_dad	0.9383	0.9425	0.282	0.278	49.26	42.91
tcn_lite_ft	0.9356	0.9395	0.272	0.266	50.15	43.62
micro_tcn	0.9354	0.9396	0.274	0.264	50.28	43.84
tcn_lite_ft_int8	0.9354	0.9393	0.269	0.263	50.18	43.74
tcn_lite	0.9352	0.9394	0.270	0.264	50.27	43.73
gru	0.9352	0.9397	0.266	0.263	50.24	43.84
tcn_lite_rsa_ws	0.9350	0.9385	0.270	0.260	50.40	43.97
mlp	0.9348	0.9390	0.265	0.263	50.63	44.30
micro_tcn_rsa_ws	0.9348	0.9384	0.272	0.257	50.67	44.44
mlp_nobp	0.9331	0.9379	0.264	0.259	51.60	45.00
linear	0.9328	0.9359	0.253	0.241	51.24	45.67
micro_tcn_rsa	0.9326	0.9373	0.263	0.260	51.87	45.37
gbdt_all	0.9321	0.9365	0.257	0.250	51.62	45.51
gbdt	0.9312	0.9359	0.248	0.244	51.99	45.85
tcn_lite_rsa	0.9311	0.9337	0.261	0.246	51.87	45.97
micro_tcn_f8	0.9299	0.9356	0.257	0.252	52.33	46.18
ema_mean	0.9289	0.9328	0.243	0.228	52.67	46.78
persistence	0.9221	0.9260	0.216	0.213	56.05	48.55
rls_ar	0.9186	0.9276	0.202	0.219	56.34	49.15
mlp_rsa	0.9159	0.9172	0.230	0.208	58.59	54.24
tcn_mha_current	0.9158	0.9218	0.189	0.179	57.45	50.11

Computed on the 16 validation and the 10 test subjects (results/validation/ and results/test/ folders), on the ENTIRE series, after dropping 80% of the beats (seed 7, from beat 0) and filling them in closed loop with the model (Burg-AR fill for the first 30 beats): **corr RR** = `np.corrcoef(original_series, modified_series)[0, 1]`; **corr dRR** = the same function on the beat-to-beat changes, `np.corrcoef(np.diff(original), np.diff(filled))[0, 1]`, which isolates the short-term dynamics. Mean over subjects; per-subject values in <folder>/crosscorr.csv. **RMSE filled**: RMSE on the model-filled beats of the same runs. Diagnostic: the selection used the round-4 protocol, not these runs.

- RR level: every model keeps a high correlation with the original (validation 0.916-0.939, test 0.917-0.942): the slow heart-rate changes dominate and the 20% kept beats anchor them. The deployed tcn_lite_ft_int8: validation 0.9354 (rank 5 of 22), test 0.9393 (rank 7); rls_ar 0.9186 / 0.9276; TCN_MHA 0.9158 / 0.9218 (validation / test).
- dRR level (beat-to-beat dynamics) separates the models more: tcn_lite_ft_int8 0.269 / 0.263, rls_ar 0.202 / 0.219, TCN_MHA 0.189 / 0.179 (validation / test); lowest: tcn_mha_current on validation, tcn_mha_current on test. Values around 0.2-0.3 are expected at 80% dropped: 4 of every 5 beat-to-beat changes are the model's own.
- Highest RR correlation: micro_tcn_dad on validation, tcn_lite_dad on test. These whole-series runs are diagnostics; the selection was made with the round-4 validation protocol and does not change.

Documents in this folder

file	content
00_overview.pdf	this summary
01_feature_engineering.pdf	the streaming feature engine, preprocessing, targets, closed-loop and int8 protocols
02_baselines.pdf	persistence, EMA mean, adaptive RLS AR(8)
03_linear.pdf	ridge regression on lagged dRR (global AR)
04_mlp.pdf	MLP on engineered features: mlp, mlp_nobp, mlp_rsa
05_gbdt.pdf	gradient-boosted trees: gbdt, gbdt_all (accuracy and feature-importance reference)
06_gru.pdf	GRU on the RR sequence + features
07_micro_tcn.pdf	separable causal TCN, 16 filters, and its variants
08_tcn_lite.pdf	full causal TCN, 24 filters, and its variants incl. the selected tcn_lite_ft
09_tcn_mha.pdf	the legacy TCN with multi-head attention (current production model)

Abbreviations and acronyms used in this document

term	meaning
AR	autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values
ar	feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model
Burg-AR	AR model whose coefficients were fitted with Burg's method (used for the first 30 beats)
Conv1D	one-dimensional convolution layer
Cortex-M7	Arm Cortex-M7, the processor core of the STM32H750 microcontroller
_dad	run-name suffix: fine-tuned with DAD (closed-loop-filled) training data
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
EMA	exponential moving average
ema_mean	baseline predicting the next RR as an exponential moving average of the window
_f8	run-name suffix: 8 convolution filters instead of 16
FLOPs	floating-point operations (arithmetic cost)
_ft	run-name suffix: fine-tuned (training continued from the parent's weights)
gbdt	gradient-boosted decision trees (run name)
GRU	gated recurrent unit (a recurrent neural-network cell)
gru	gated recurrent unit (run name)
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
KB	kilobyte (1,024 bytes)
MAC	multiply-accumulate operation (one multiplication plus one addition): the unit of compute cost on the microcontroller; MACs/beat = how many the model needs per heartbeat
MACs	multiply-accumulate operations (see MAC)
MCU	microcontroller unit
MHA	multi-head attention (a transformer layer, not supported by the MCU toolchain)
MLP	multilayer perceptron (a fully connected neural network)
mlp	multilayer perceptron (run name)
ms	milliseconds
nobp	run-name part: no band-pass (the fixed RSA band-pass features bp and bp_prev removed)
np.corrcoef	NumPy's Pearson correlation-coefficient function
np.diff	NumPy's successive-difference function
ReLU	rectified linear unit, $\max(0, x)$ (an activation function)
RLS	recursive least squares: an adaptive filter that updates its coefficients at every beat
rls_ar	baseline: adaptive per-patient AR(8) forecast by RLS
RMSE	root mean square error (ms): the main accuracy measure, lower is better
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
RSA	respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing
_rsa	run-name suffix: uses the RSA-tracker features
SeparableConv1D	depthwise-separable 1-D convolution: a per-channel convolution followed by a pointwise one (much cheaper than Conv1D)
STM32H750	STMicroelectronics 32-bit microcontroller (Arm Cortex-M7, 128 KB flash, 1 MB RAM)
TCN	temporal convolutional network (causal, dilated 1-D convolutions)
tcn	temporal convolutional network (run-name part)
val	validation: the 16 validation subjects, used for model selection
_ws	run-name suffix: warm start (initialised from the parent's weights, new inputs set to zero)